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Carbon Coated Fe₃O₄ Nanospindles as a Superior Anode Material for Lithium-Ion Batteries**

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Carbon-coated Fe₃O₄ nanospindles are synthesized by partial reduction of monodispersed hematite nanospindles with carbon coatings, and investigated with scanning electron microscopy, transmission electron microscopy, X-ray diffraction, and electrochemical experiments. The Fe₃O₄-C nanospindles show high reversible capacity ($\sim 745 \text{ mA h g}^{-1}$ at C/5 and $\sim 600 \text{ mA h g}^{-1}$ at C/2), high coulombic efficiency in the first cycle, as well as significantly enhanced cycling performance and high rate capability compared with bare hematite spindles and commercial magnetite particles. The improvements can be attributed to the uniform and continuous carbon coating layers, which have several functions, including: i) maintaining the integrity of particles, ii) increasing the electronic conductivity of electrodes leading to the formation of uniform and thin solid electrolyte interphase (SEI) films on the surface, and iii) stabilizing the as-formed SEI films. The results give clear evidence of the utility of carbon coatings to improve the electrochemical performance of nanostructured transition metal oxides as superior anode materials for lithium-ion batteries.

1. Introduction

Rechargeable lithium-ion batteries have been the most utilized batteries in the portable electronic market for many years. They have the highest energy density available from existing rechargeable battery techniques, but their performance still lies behind the demands of the consumer. New electrode materials with high specific capacities are necessary to meet these demands. Recently, transition metal oxides have shown desirable properties, such as high theoretical capacity ($\sim 500\text{--}1000 \text{ mA h g}^{-1}$, compared with $\sim 372 \text{ mA h g}^{-1}$ for conventional graphite), based on a novel conversion mechanism,^[1] making them promising anode materials for high performance lithium-ion batteries.^[1–10] However, most of the transition metal oxides usually suffer from the problem of poor electronic conduction, and hence need to be modified by admixing electronically conductive phases.^[11] There are very

few transition metal oxides such as Fe₃O₄ and RuO₂ with high electronic conductivities.^[6–8,12,13] As Fe₃O₄ has advantages over RuO₂ in terms of low cost and environmental benignity, extensive research work has focused on it recently.^[6–8] Another challenge in using transition metal oxides as anode materials is their poor cycling performance; the electrodes cannot maintain their integrity over several discharge/charge cycles. It has been widely demonstrated that this problem can be partly solved in nanostructured electrode materials because of better accommodation of the strains of Li insertion/extraction than in micrometer-scale materials.^[1,14–19] However, the high surface area of nanostructured electrode materials raises the risk of secondary reactions involving electrolyte decomposition between electrode and electrolyte, which causes a high level of irreversibility (i.e., low coulombic efficiency) and poor life cycle, and the formation of thick solid electrolyte interphase (SEI) films on the electrode surface, which consume much of the Li supplied by the cathodes.^[12] Furthermore, it has been found in many nanostructured transition metal oxides that thick SEI films formed during Li uptake may disappear completely, catalyzed by transition metal upon Li extraction, which leads to capacity fading and safety problems.^[1,9,13,20,21] Therefore, surface modifications must be considered for nanostructured transition metal oxides.

Carbon coating is one of the most widely used surface modification techniques for cathode materials (e.g., LiFePO₄), because carbon coating layers can significantly enhance the electronic conductivity of electrode materials, which results in improved rate performance.^[22–24] In fact, carbon materials are very stable anode materials in lithium-ion batteries due to the small volume change during Li insertion/extraction. The SEI films on carbon surface are also relatively stable.^[21,25,26] Therefore, carbon coatings should be also used for anode materials since they may serve as perfect barriers to protect the

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inner active materials and maintain their high capacities.^[26–29] In this Full Paper, we successfully synthesize a superior anode material for lithium-ion batteries by introducing carbon coatings on the surface of iron oxide nanospindles. It is found that the carbon coating layers not only enhance the electronic conductivity of electrode materials, but also lead to stabilized SEI films. As a result, the as-prepared carbon coated Fe_3O_4 nanospindles exhibit very high specific capacity ($\sim 749 \text{ mA h g}^{-1}$ at C/5 and $\sim 600 \text{ mA h g}^{-1}$ at C/2), significantly improved cycling performance, and high rate performance compared with commercial magnetite particles and bare hematite nanospindles.

2. Results and Discussion

The carbon coated magnetite nanospindles were prepared by in situ partial reduction of hematite spindles with carbon coatings. First, the monodispersed spindle hematite particles were synthesized by forced hydrolysis of ferric chloride solutions with small additions of phosphate ions at elevated temperatures.^[30] Figure 1a shows the scanning electron microscopy (SEM) image of as-synthesized hematite spindles. All particles are spindles alike, composed of tiny crystals. They have a mean particle length of about 470 nm and a diameter of about 106 nm. The detailed statistical distributions of the particle length and diameter are shown in Figure 2a and b, respectively.

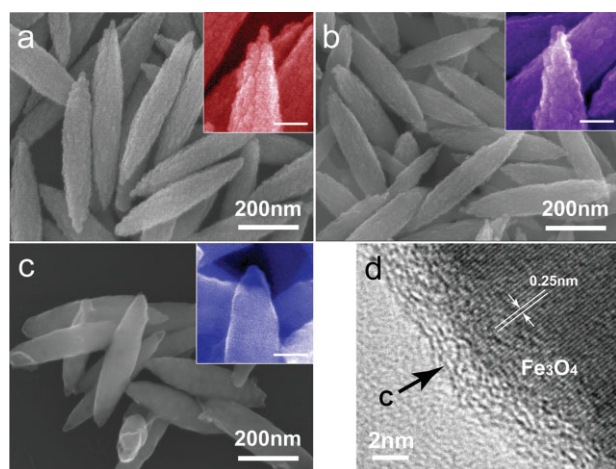


Figure 1. a) SEM image of the as-synthesized hematite spindles. b) SEM image of the carbon precursor coated hematite spindles. c) SEM image of the carbon coated Fe_3O_4 spindles (Fe_3O_4 -C composites). d) high-resolution transmission electron microscopy (TEM) image of the Fe_3O_4 -C composites. The insets are close views of corresponding samples, all unmarked scale bars are 50 nm.

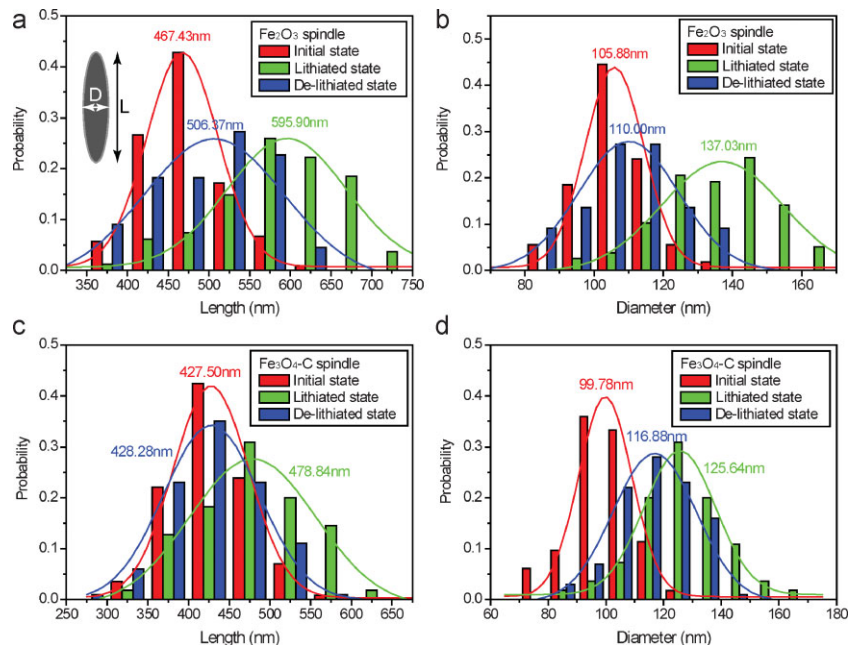


Figure 2. The a) length and b) diameter distributions of bare $\alpha\text{-Fe}_2\text{O}_3$ nanospindles in initial, lithiated and de-lithiated state; the c) length and d) diameter distributions of Fe_3O_4 -C nanospindles in initial, lithiated, and de-lithiated state. The inset in the left top picture shows the meaning of length and diameter for a spindle.

Figure 3a shows the power X-ray diffraction (XRD) pattern of the same sample, in which all peaks are in good agreement with $\alpha\text{-Fe}_2\text{O}_3$ (JCPDS no. 89-8103, $a = b = 5.020 \text{ \AA}$, $c = 13.719 \text{ \AA}$). Both the morphology and structure are in agreement with those reported in the literature.^[30]

Secondly, the carbon precursor layers were coated on the outer surface of the hematite spindles by pyrolysis of glucose under hydrothermal conditions.^[29,31] Figure 1b shows the SEM micrograph of the carbon-precursor-coated hematite spindles; it can be seen that there is no significant change in morphology, indicating uniform carbon coatings. Finally, the products were dried and heat-treated at 600°C for about 12 h under N_2

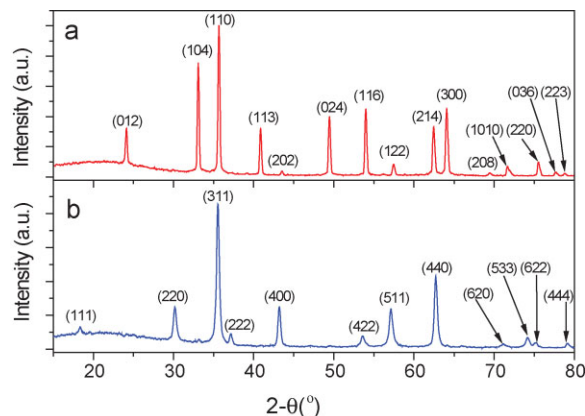


Figure 3. XRD patterns of a) hematite spindles, and b) Fe_3O_4 -C composites. They are identified as rhomb-centered $\alpha\text{-Fe}_2\text{O}_3$ (JCPDS no. 89-8103) and face-centered Fe_3O_4 (JCPDS no. 65-3107), respectively.

atmosphere to carbonize the carbon precursor layers. At the same time, the inner hematite spindles were partially reduced to magnetite by the outer carbon layers, hence leading to the final $\text{Fe}_3\text{O}_4\text{-C}$ nanocomposites. Figure 1c shows the SEM micrograph of the final composites; they have much smoother surfaces than the precursor. The mean length and diameter of the resultant $\text{Fe}_3\text{O}_4\text{-C}$ spindles were about 428 and 100 nm, respectively (see Figure 2c and d for details). It can also be seen, from the inset in Figure 1c, that the composites have a small void space at the tips of the spindles. All evidence indicates that there is slight volume shrinkage during the heat-treatment process, which is likely to be due to the partial reduction process of $\alpha\text{-Fe}_2\text{O}_3$. The high-resolution transmission electron microscopy (TEM) micrograph of the composite is shown in Figure 1d; both the outer carbon layer and the inner magnetite core are visible. The carbon layer, which has a thickness of $\sim 2\text{--}10\text{ nm}$, is uniform and continuous. More detailed TEM micrographs of the carbon layers can be found in Figure S1 of the Supporting Information.

To clarify the structure of the composites, XRD experiments were carried out. Figure 3b shows the XRD pattern of the final $\text{Fe}_3\text{O}_4\text{-C}$ spindles, which is in good agreement with face-centered Fe_3O_4 (JCPDS no. 65-3107, $a = b = c = 8.390\text{ \AA}$). No obvious XRD peaks corresponding to graphite are found in the XRD pattern, which indicates the carbon coatings are not well crystallized. The $\gamma\text{-Fe}_2\text{O}_3$ particles have similar XRD patterns to Fe_3O_4 , but they cannot exist at the heat-treatment temperature as they convert to $\alpha\text{-Fe}_2\text{O}_3$ at about 400°C . Therefore, the products can be identified with Fe_3O_4 , which is further confirmed by the color change from brown to totally black during heat treatment and the strong ferromagnetism of the final black products. It can be concluded that the $\alpha\text{-Fe}_2\text{O}_3$ spindles were fully converted to Fe_3O_4 because the peaks unique for Fe_2O_3 (including the peaks at 24.1 , 33.08 , and 40.86° in Figure 3a), disappeared completely in Figure 3b. Note that the slightly broadened XRD peaks occurring after the partial reduction process are ascribed to the crystal size change from about 40 nm in hematite spindles to 30 nm in magnetite spindles, indicating that both of the spindles are composed of fine crystalline particles.

In order to investigate the pore structure of the $\alpha\text{-Fe}_2\text{O}_3$ and $\text{Fe}_3\text{O}_4\text{-C}$ spindles, a nitrogen isothermal adsorption technique was used. The results are shown in Figure S2 of the Supporting Information. The Brunauer–Emmett–Teller (BET) surface areas of the $\alpha\text{-Fe}_2\text{O}_3$ and $\text{Fe}_3\text{O}_4\text{-C}$ spindles are 14.6 and $35.1\text{ m}^2\text{ g}^{-1}$, respectively. The Barrett–Joyner–Halenda (BJH) pore size distribution indicates that, besides the big interspaced pores among particles, the composite has mesopores of about 3.8 nm compared with bare Fe_2O_3 , which may also come from the crystal volume shrinkage during the partial reducing process. They are helpful for buffering the volume expansion during lithium insertion, which may contribute to a better cycling performance. Furthermore, inductively coupled plasma-optical emission spectroscopy was used to reveal the precise chemical composition of the final composites. The results showed that the composites have a chemical composi-

tion of $78.5\text{ wt\% Fe}_3\text{O}_4$ and $21.5\text{ wt\% carbon}$, which leads to a theoretical specific capacity of 805 mA h g^{-1} for the $\text{Fe}_3\text{O}_4\text{-C}$ spindles when being used as anode material in lithium-ion batteries. Note that we include the mass of carbon coating layers when calculating the specific capacity of $\text{Fe}_3\text{O}_4\text{-C}$ composite in the context.

In order to test the applicability of the new composites in lithium-ion batteries, the electrochemical properties with respect to Li insertion/extraction were investigated. The electrochemical properties of bare $\alpha\text{-Fe}_2\text{O}_3$ spindles and commercial Fe_3O_4 particles (Alfa Aesar, $\sim 300\text{ nm}$ in diameter) were also investigated for comparison. Although the commercial magnetite particles and the $\text{Fe}_3\text{O}_4\text{-C}$ spindles have different morphology, they have comparable particle size. The SEM images of the commercial Fe_3O_4 particles, $\alpha\text{-Fe}_2\text{O}_3$ spindles, and $\text{Fe}_3\text{O}_4\text{-C}$ composites on copper foil electrodes are shown in Figure S3, S4a, and S4d of the Supporting Information, respectively. The first five discharge/charge cycles were tested at a relatively low current rate of $C/5$ ($8\text{ lithium per Fe}_3\text{O}_4$ in 5 h), and the corresponding discharge/charge voltage profiles are shown in Figure 4. In the first discharge curve, that of the bare $\alpha\text{-Fe}_2\text{O}_3$ (Fig. 4a), there are two obvious potential plateaus at about 1.6 and 0.8 V versus Li^+/Li , which are similar to those reported for Fe_2O_3 .^[4,32] In the cases of the commercial Fe_3O_4 (Fig. 4b) and the $\text{Fe}_3\text{O}_4\text{-C}$ composites (Fig. 4c), the first discharge curves are very similar, with only one potential plateau at about 0.8 V versus Li^+/Li , which are close to that described in the literature for Fe_3O_4 anodes.^[7,8] The specific charge capacities of both the commercial Fe_3O_4 particles and the bare $\alpha\text{-Fe}_2\text{O}_3$ quickly fade, and show poor cycling performance, while the $\text{Fe}_3\text{O}_4\text{-C}$ composite shows much better performance. The first specific charge capacity of $\text{Fe}_3\text{O}_4\text{-C}$ composite is as high as 749 mA h g^{-1} , approaching

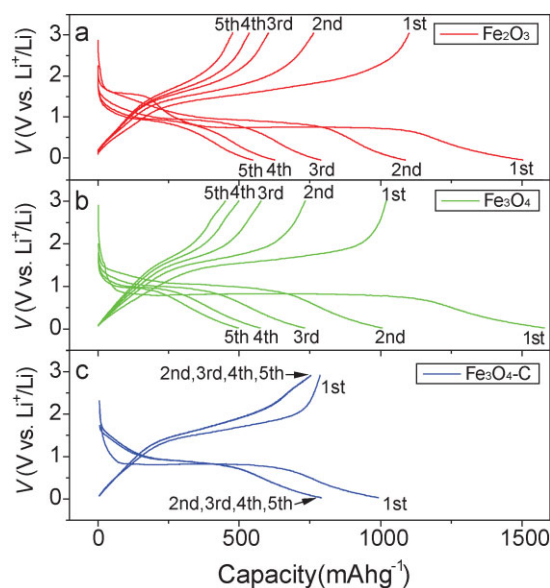


Figure 4. The discharge/charge profiles of a) bare hematite spindles, b) commercial magnetite particles, and c) $\text{Fe}_3\text{O}_4\text{-C}$ composites, at a current rate of $C/5$.

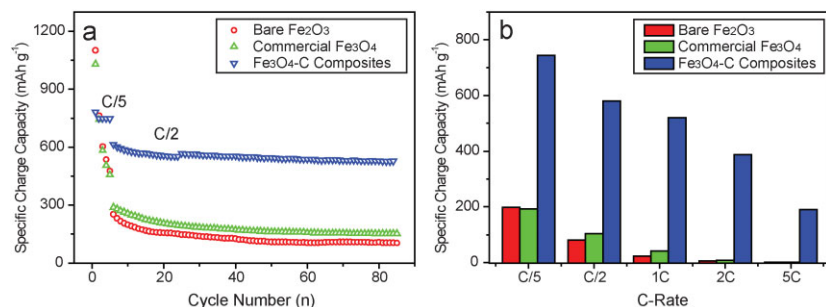


Figure 5. a) Cycle performance and b) rate performance of the bare α -Fe₂O₃ spindles, commercial Fe₃O₄ particles and Fe₃O₄-C composites.

~93% of its theoretical specific capacity (805 mA h g⁻¹). The capacity of the composite remains unchanged through the subsequent four discharge/charge cycles, as shown in Figure 4c, in which the discharge/charge curves of these cycles overlap. The initial coulombic efficiency of the Fe₃O₄-C composite is ~80%, which is significantly higher than ~70% of the bare α -Fe₂O₃ spindles and ~65% of the commercial Fe₃O₄ particles, showing a remarkable decrease of the irreversible capacity loss. The subsequent 80 cycles were tested at a relatively high current rate of C/2. The results from these cycles are shown in Figure 5a. It was found that the capacity of the bare α -Fe₂O₃ and the commercial Fe₃O₄ fade to 250 and 300 mA h g⁻¹ immediately and then gradually decreased to 105 and 152 mA h g⁻¹, respectively. The capacity of Fe₃O₄-C composites also decrease to about 600 mA h g⁻¹, but the capacity remained 530 mA h g⁻¹ even after another 80 discharge/charge cycles, which is almost 5 times of that of the bare α -Fe₂O₃ spindles and 3.5 times of that of the commercial Fe₃O₄ particles.

In addition to the much improved cycling performance, the Fe₃O₄-C composites also show significantly enhanced high rate performance compared with the bare α -Fe₂O₃ spindles and the commercial Fe₃O₄ particles (Fig. 5b). At a high rate of 5 C (discharge/charge of all active materials within 12 min, i.e., 4620 mA g⁻¹), the specific charge capacity of Fe₃O₄-C composites is still 190 mA h g⁻¹, whereas α -Fe₂O₃ spindles and commercial Fe₃O₄ particles nearly have no capacities at this condition. The results indicate that the improvement in electrochemical performance, in terms of cycling performance and rate performance, should not be attributed to the magnetite itself or the spindle type morphology, but the carbon coating layers outside the magnetite.

In order to well understand the effects of the carbon coatings on the electrochemical properties of Fe₃O₄ nanospindles, the intermediate states of the Fe₃O₄-C composites during Li insertion/extraction were investigated in detail. The anode materials were investigated on the electrodes by SEM. Figure S4 of the Supporting Information shows the SEM images of the electrodes of bare α -Fe₂O₃ spindles and Fe₃O₄-C composites in the initial, lithiated and de-lithiated state. Since the Fe₃O₄-C composites have continuous carbon coatings (see Figure S1 of the Supporting Information), they can form a more efficient electron conductive network, which is helpful

for both cycle performance and rate performance.^[11] In addition, nearly all Fe₃O₄-C spindles are unbroken on the battery electrode, indicating the composites are stable and strong enough to withstand the casting and pressing processes necessary for battery fabrication. The α -Fe₂O₃ and Fe₃O₄-C anode materials were further investigated by TEM after several discharge/charge cycles. Figure 6 shows the TEM images of the α -Fe₂O₃ and Fe₃O₄-C spindles in both the fully lithiated (discharge) and fully de-lithiated (charge) states. The spindle-like morphology of both materials remained

unchanged during the discharge/charge cycles in both states. For the Fe₃O₄-C composites, as shown in Figure 6a and b, the total thickness of the outer carbon layer plus the SEI film is relatively thin, which is about 10 nm in fully lithiated and de-lithiated states. Furthermore, the thickness of the SEI films has no significant change during the insertion/extraction of lithium. However, the SEI films on the bare α -Fe₂O₃ spindles are much more thick and unstable, as shown in Figure 6c and d, which can be 20–50 nm in fully lithiated state while 0–20 nm in de-lithiated state. Similar reports have been found on other bare transition metal oxides such as RuO₂^[12], Cr₂O₃^[21], and Co₃O₄^[26]. To further confirm the results, the length and the diameter distributions of both spindle-like anode materials in the initial, lithiated and de-lithiated states were analyzed based on the SEM images. Note that we include the lengths and diameters of the SEI films when analyzing the data. The statistical results are shown in Figure 2. All data in this figure come from corresponding SEM images similar to Figure S4 of the Supporting Information but with a magnification of 20000 $\times$, and the sample capacities are about 100 in all statistics. From the results, we can see that the length change and diameter change of the bare α -Fe₂O₃ spindles between the

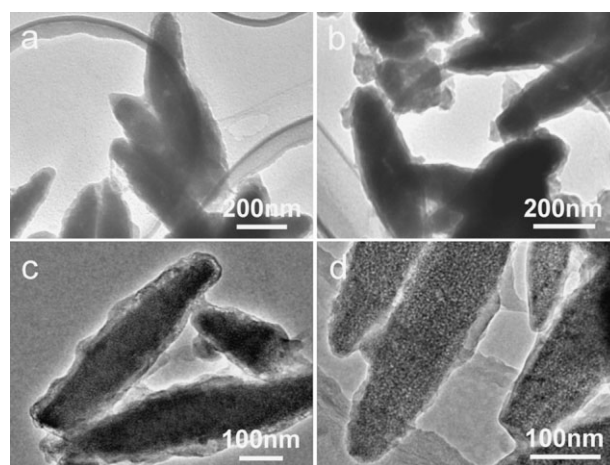


Figure 6. a) TEM images of the Fe₃O₄-C composites in a fully lithiated state. b) Fe₃O₄-C composites in a fully de-lithiated state. c) Bare α -Fe₂O₃ composites in a fully lithiated state. d) Bare α -Fe₂O₃ composites in a fully de-lithiated state after 3 discharge/charge cycles at a rate of C/5.

lithiated and de-lithiated states is about 90 nm (596 nm – 506 nm) and 27 nm (137 nm – 110 nm), respectively. However, in the case of Fe_3O_4 -C spindles, the length change is only about 51 nm (479 nm – 428 nm), and the diameter change is only 9 nm (126 nm – 117 nm). This means that the SEI films are much more stable on the composite surface, namely under much smaller volume changes. The results indicate that the outer carbon layer can protect the inner active material as well as restrain the formation/decomposition of the SEI films. Note that even in the absence of carbon black, the Fe_3O_4 -C itself still performs better than the bare α - Fe_2O_3 and the commercial Fe_3O_4 without carbon black. These results indicate that it is the outer carbon layer that contributes most to the great enhancement of both the cycle performance and the rate performance.

3. Conclusions

In summary, we synthesized new carbon coated Fe_3O_4 spindles with length of about 500 nm and an axis ratio of about 4. It was demonstrated that outer carbon layers with uniform and continuous features improve the electrochemical performance in several ways, including: i) maintaining the integrity of particles, ii) increasing the electronic conductivity of electrodes leading to the formation of uniform and thin SEI films on the surface, and iii) stabilizing the as-formed SEI films. As a result, a Fe_3O_4 -C composite is a promising anode material for highly efficient lithium-ion batteries with high reversible capacity ($\sim 745 \text{ mA h g}^{-1}$ at C/5 and $\sim 600 \text{ mA h g}^{-1}$ at C/2), high coulombic efficiency in the first cycle, and significantly enhanced cycling performance and high rate capability compared with bare hematite spindles and commercial magnetite particles. The results presented here give clear evidence of the ability of carbon coatings to improve the electrochemical performance of nanostructured transition metal oxides as superior anode materials for lithium-ion batteries. We believe that the strategy will pave the way for the practical application of nanostructured transition metal oxides and, hence, make possibly the next generation of high-energy-density lithium-ion batteries.

4. Experimental

Preparation of Hematite Nanospindles: All reagents below were received from Beijing Chemicals Co. (Beijing, China) and used without further purification. A 75 mL mix solution containing 0.02 M $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and 0.45 M NaH_2PO_4 in deionized (DI) water was transferred to a 100 mL teflon autoclave, and then heated at 105 °C in an electric oven for 48 h. After the autoclave cooled down, the monodispersed hematite particles were centrifuged and washed with DI water.

Preparation of Carbon Coated Fe_3O_4 Nanospindles: A 300 mg hematite particles was dispersed in 5 mL DI water by ultrasonication to form a suspension. 600 mg glucose ($\text{C}_6\text{H}_{12}\text{O}_6 \cdot 3\text{H}_2\text{O}$) was dissolved in another 15 mL DI water, and the former suspension and another 10 mL of ethanol were added to the solution under gentle stir. The resulting suspension was transferred to a 40 mL Teflon autoclave, which was then heated at 190 °C in an electric oven for 12–15 h. The carbon-precursor-coated hematite particles were harvested by centrifugation

and washed with DI water, then dried at 110 °C in an oven. The resulting sample was heated in a quartz tube at a rate of 6 °C min^{-1} to 600 °C in N_2 atmosphere. The temperature was kept constant for another 12 h before the tube was allowed to cool down to room temperature.

Inductively Coupled Plasma-Optical Emission Spectroscopy Measurement: 13.3 mg dried Fe_3O_4 -C composites were completely dissolved in 5 mL concentrated nitric acid by heating, and the resulting yellow solution was diluted to 1000 mL with DI water. The resulting solution was analyzed by Optima 5300DV, and the concentration of the Fe element was 7.556 mg L^{-1} .

Characterization: The products were characterized by SEM (JEOL 6701F), TEM (TEM, JEM JEOL 2010 and 1011), XRD (Rigaku Dmax-rB diffractometer with $\text{CuK}\alpha$ radiation, $k = 0.1542 \text{ nm}$, 40 kV, 100 mA). All SEM and TEM samples of active material in the lithiated/de-lithiated state were prepared in an argon-filled glove box, washed three times with dimethyl carbonate (DMC), and sealed in an argon-filled jar before characterization.

Electrochemical Characterization: Electrochemical experiments were performed using Swagelok-type cells. The working electrodes were prepared by mixing the α - Fe_2O_3 spindles, commercial Fe_3O_4 particles or Fe_3O_4 -C composites, carbon black, and poly(vinylidene fluoride) (PVDF) at a weight ratio of 70:15:15, and pasted on pure Cu foil. Glass fiber (GF/D) from Whatman was used as a separator. The electrolyte consists of a solution of 1 M LiPF_6 in ethylene carbonate (EC)/dimethyl carbonate (DMC)/diethyl carbonate (DEC) (1:1:1, in wt%) obtained from Ferro Corporation. Pure lithium foil was used as counter electrode. The discharge and charge measurements were carried on an Arbin BT2000 system. The cells were assembled in an argon-filled glove box.

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